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"Tool-in-lesion" Accuracy of Galaxy System—A Robotic Electromagnetic Navigation BroncHoscopy With Integrated Tool-in-lesion-Tomosynthesis Technology

The MATCH Study

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Background: **The Galaxy System (Noah Medical) is a novel robotic endoluminal platform using electromagnetic navigation combined with integrated tomosynthesis technology and augmented fluoroscopy. It provides intraprocedural imaging to correct computerized tomography (CT) to body divergence and novel confirmation of tool-in-lesion (TIL). The primary aim of this study was to assess the TIL accuracy of the robotic bronchoscope with integrated digital tomosynthesis and augmented fluoroscopy.**

Methods: **Four operators conducted the experiment using 4 pigs. Each physician performed between 4 and 6 nodule biopsies for 20 simulated lung nodules with purple dye and a radio pacifier. Using Galaxy's "Tool-in-Lesion Tomography (TOMO+)" with augmented fluoroscopy, the physician navigated to the lung nodules, and a tool (needle) was placed into the lesion. TIL was defined by the needle in the lesion determined by cone-beam CT.**

Results: **The lung nodule's average size was 16.3 ± 0.97 mm and was predominantly in the lower lobes (65%). All 4 operators successfully navigated to all (100%) of the lesions in an average of 3 minutes and 39 seconds. The median number of tomosynthesis sweeps was 3 and augmented fluoroscopy was utilized in most cases (17/20 or 85%). TIL after the final TOMO sweep was 95% (19/20) and tool-touch-lesion was 5% (1/20). Biopsy yielding purple pigmentation was also 100% (20/20).**

Conclusion: **The Galaxy System demonstrated successful digital TOMO confirmed TIL success in 95% (19/20) of lesions and tool-touch-lesion in 5% (1/20) as confirmed by cone-beam CT. Successful diagnostic yield was achieved in 100% (20/20) of lesions as confirmed by intralesional pigment acquisition.**

Key Words: navigational bronchoscopy, electromagnetic geopositional navigation, lung nodule, digital tomosynthesis, cone-beam CT bronchoscopy

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Lung cancer remains the deadliest form of cancer, with over 150,000 deaths per year.[1] An estimated 1.5 million lung nodules are detected each year.[2] There are only 2 commonly accepted minimally invasive procedures used to biopsy peripheral lung nodules: (1) transthoracic needle aspiration (TTNA) and (2) advanced diagnostic bronchoscopy. Compared with computerized tomography (CT)-guided TTNA (CT-TTNA), advanced diagnostic bronchoscopy has a better safety profile (lower risk of pneumothorax, life-threatening bleeding, and length of stay) and the ability to stage the mediastinum in one procedure. Unfortunately, the diagnostic yield of bronchoscopy is lower than CT-guided TTNA.[3] Traditional advanced diagnostic bronchoscopy systems have a limited diagnostic yield of 70% to 73% for diagnosing peripheral lung nodules.[4]

The advent of robotic navigational platforms has garnered renewed interest in the biopsy of peripheral lung lesions. Robotic platforms offer superior stability, distal articulation, and visualization over traditional precurved catheters.

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Current robotic platforms (Ion, Intuitive Surgical and Monarch, Auris Surgical Robotics) respectively utilize either shape sensing (SS) technology with an embedded fiber optic sensor that measures the shape of the catheter several hundred times per minute or electromagnetic navigation (EMN) combined with insertion distance and image/airway recognition for guidance.[5]

Both SS and EMN bronchoscopy systems are thought to be prone to CT-to-body divergence (CT2BD). CT2BD is the discrepancy between the electronic virtual target and the actual anatomic location of the peripheral lung lesion.[6] A myriad of factors is thought to cause CT2BD including differences in lung volumes at the time of the preprocedural scan and bronchoscopy. Variations with respiration can lead to movement of the target lesion on average up to 17.6 mm—sometimes larger than the lesion itself.[7] To overcome CT2BD, many bronchoscopists are supplementing advanced imaging devices such as digital tomosynthesis, cone beams, and *O*-arms.[8] Pritchett[9] reported a total of 93 lesions with a median size of 16.0 mm were biopsied in 75 consecutive patients with a diagnostic yield of 83.7% based on the strict AQuIRE definition. Although cone-beam CT is generally accepted as the gold standard of intraoperative imaging, access and cost of fixed and mobile cone-beam CT systems limit widespread adoption.[10]

The largest prospective single-arm cohort study of navigational bronchoscopy, NAVIGATE, demonstrated a diagnostic yield of 67.4%.[11] The advent of EMN bronchoscopy with digital tomosynthesis, DT-ENB, (SuperDimension Medtronic) promised an improved diagnostic yield over previous legacy systems. Katsis et al[3] demonstrated using DT-ENB in 363 peripheral lung lesions and achieved a diagnostic yield of 77.4%.

In a recent single-center study, the diagnostic yield of SS versus DT-ENB was 77% (110/143 Peripheral lung lesions) and 80% (158/197 Peripheral lung lesions) ($P = 0.4$).[12] Although this study suggests SS and DT-ENB are comparable and offer a stepwise improvement over previous platforms, there is still significant room for improvement in the diagnostic yield. Combining robotic bronchoscopy with integrated intraprocedural image guidance may offer an improvement in diagnostic yield over current robotic and DT-ENB platforms.

The Galaxy System (Noah Medical) is a robotic endoluminal platform that combines EMN with integrated tomosynthesis technology and augmented fluoroscopy. (Fig. 1) The Galaxy System is designed to utilize the advantages of robotic bronchoscopy and mitigate CT2BD with a novel confirmation of tool-in-lesion (TIL). TIL is defined as the biopsy needle in the lesion. The primary aim of the "Tool-in-lesion" accuracy of Galaxy System, a robotic electromagnetic navigation bronchoscopy, with integrated TIL-tomosynthesis (TILT) technology, the MATCH Study, was to assess the TIL accuracy of the robotic bronchoscope with integrated digital tomosynthesis and augmented fluoroscopy as confirmed by cone-beam CT imaging.

METHODS

Animal Preparation

A porcine model (Sus scrofa domesticus) was utilized. Pig lung anatomy has significant similarities to the anatomy and physiology of the human lung and is deemed an appropriate animal model for many bronchoscopy trials.[13] The study was approved by the Sutter Institute for Medical Research Institutional Animal Care and Use Committee Protocol NRE.02.22. Animal husbandry, preparation, and euthanasia were performed according to accepted ethical and Institutional Animal Care and Use Committee guidelines.

Each pig was anesthetized with volatile gas and underwent tracheostomy with an 8.5 mm endotracheal tube and bilateral chest tube thoracostomy. Anesthesia was monitored by a veterinarian with invasive hemodynamic monitoring.

Under fluoroscopic guidance, simulated lung nodules were created by percutaneous injection of a gelatinous solution containing purple-colored



FIGURE 1. Image of the robotic endoluminal navigation platform, Galaxy System. Image provided by Noah Medical (Noah Medical).

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tracer material and a radio pacifier into the lung periphery. A CT was then performed with a breath hold at inspiration with an adjustable pressure limit set to 25 cm H_2O for preprocedure planning. All injected lung nodule targets were peripheral lesions in that they were surrounded by normally aerated lung, none were endobronchial, and all were beyond the segmental bronchus so that all biopsies were transbronchial rather than endobronchial. All lesions were at least 8 mm in the largest diameter.

TIL was defined by the needle in the lesion. Tool-touch-lesion (TTL) was defined as a needle that is tangential or touching the lesion but is not within the lesion. The center strike was defined as the needle in the middle third in 3 orthogonal planes (axial, sagittal, and coronal). TIL definitions can be seen in Figure 2 and Figure 3, for example, visualizations on preplanning CTs, tomography (TOMO), and cone-beam CT (CBCT) images. Nodule size was calculated using the average of the longest and shortest dimensions on the preplanning CT scan of the chest. This scan was also used to determine whether a bronchus sign was present and categorize the nodule's location (middle or peripheral lung zone).

Anesthesia Protocol

After tracheostomy, lung recruitment was performed by giving 4 positive end-expiratory pressure (PEEP) recruitment breaths (30 cm H_20 over 30 s). Repeated recruitment maneuvers were performed if atelectasis was noted on the helical CT. Pigs were ventilated using volume control ventilation with a high tidal volume (8 to 12 mL/kg) and a high PEEP strategy set at 15 cm H_20. An apneic breath hold was performed during image acquisition and the adjustable pressure-limiting valve was set to 25 cm H_20. If apneic breath-hold strategies were used, apnea was maintained no longer than 2 minutes. Continuous end-tidal CO_2 monitoring was utilized as a safety measure. Each pig underwent between 4 and 6 navigational bronchoscopies and multiple scans with a portable *C*-arm and fixed cone-beam CT imaging. The trial conditions were much longer than a typical human navigational procedure and the I-LOCATE study demonstrated that one of the major factors of atelectasis is the time of the procedure.[14] Given the prolonged nature of the study, we utilized a modified version of a published lung navigation ventilation protocol with high tidal volume, high PEEP, and breath-hold strategies to mitigate atelectasis.[10]

Navigation

Over 4 separate days, 4 operators (the authors) conducted the experiment using 4 pigs. Each physician performed 6, 5, 5, and 4 nodule biopsies, respectively, for a total of 20 lung nodule biopsies.

Galaxy Planning Software was used to identify and mark target lesions on the CTs as well as plan pathways on the segmented airway tree. The robotic platform was set up, airway registration was performed, and an individual target lesion was selected. The bronchoscope was guided to the desired target lesion using a handheld controller under electromagnetic guidance (Fig. 4; Galaxy System User Interface). A tomosynthesis sweep was performed utilizing a 2-dimensional fluoroscopic c-arm with a 9 inch image intensifier (OEC 9900 Elite, GE). The *C*-arm sweep consisted of a limited circular rotation of 30 degrees left anterior oblique to 30 degrees right anterior oblique. The bronchoscope tip was marked in the software. Based on the reconstruction algorithm, 2-dimensional images were stacked to create a section image, on which the target nodule was marked.

The bronchoscope was then navigated to the corrected target and a needle was placed. If desired, the operator could utilize augmented fluoroscopy to help optimize the bronchoscope and tool position. A repeat tomosynthesis sweep was performed to confirm the TIL confirmation. Repeated attempts were allowed at the user's discretion until

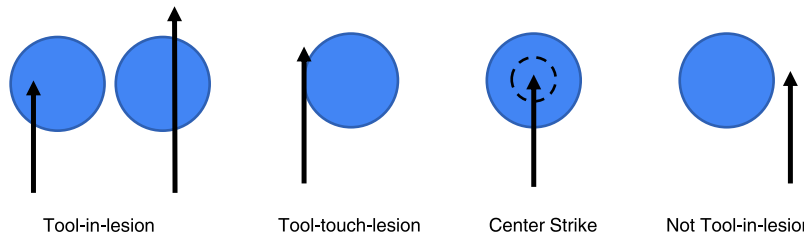


FIGURE 2. Tool in the lesion was defined as the biopsy needle in or tangential to the lesion.

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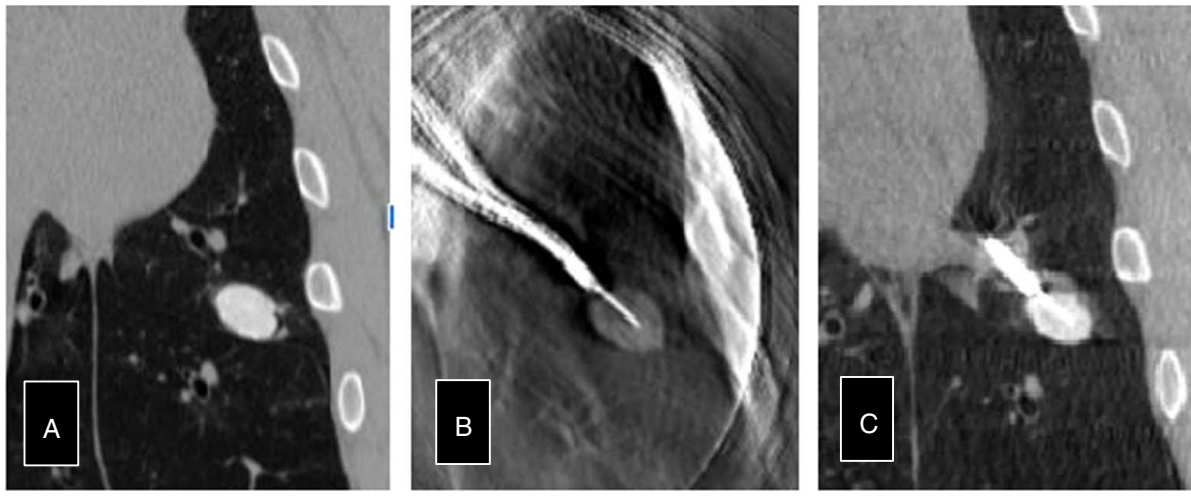


FIGURE 3. Example of a left upper lobe lung nodule. Visualization of lesions on the preprocedural spin (A), tool-in-lesion (TIL)-tomosynthesis (TILT+) TIL confirmation (B), and final cone-beam computerized tomography (CBCT) confirmation (C).

the needle was optimally positioned. The digital tomosynthesis TIL confirmation was based on the "TOMO reconstruction coordinate technique" described below.

Once the final position was confirmed, a CBCT scan was captured with an 8-second sweep (0.5 projection/degree, 396 projections). An 8-second spin was performed to optimize image quality. CBCT TIL confirmation was defined as needle placement either in the lesion or TTL defined as tangential to the lesion in three orthogonal planes (axial, sagittal, and coronal). The catheter position was not adjusted after the confirmatory CBCT scan (Fig. 5; study workflow). The time to navigation was determined from the start of navigation to the time that biopsies were concluded. Radiation exposure was recorded in milligray (mGy). The number of tomosynthesis sweeps and fluoroscopy time were recorded. After confirmation with CBCT, needle aspiration was performed. Needle passes were diagnostic if purple pigment was visualized on gross inspection or microscopic evaluation.

Tomography Reconstruction Coordinate Technique

A digital tomosynthesis sweep was performed as described. The reconstructed tomosynthesis image consists of a coordinate representing the depth of the displayed slice within the reconstruction in the anterior-posterior direction. The distance between the optimal image slice of the needle and the optimal image of the lesion was measured. The coordinate feature can help inform whether the tool is in the lesion by calculating the difference between the coordinates for the depths of the needle and the lesion, representing the distance between the needle and the

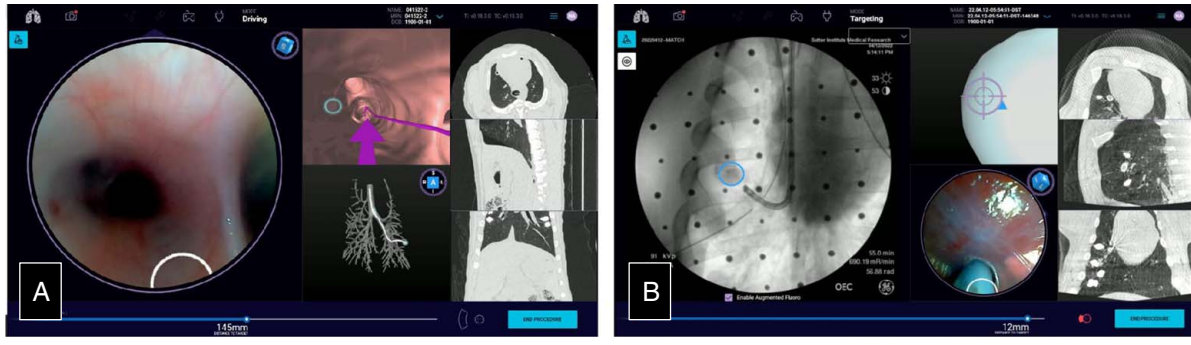


FIGURE 4. Galaxy System User Interface. A, The screenshot on the left demonstrates the endoluminal view and virtual pathways to the lesion and axial, sagittal, and coronal views on the preplanning computerized tomography (CT) of the chest. B, The screenshot on the right demonstrates an augmented fluoroscopy outline of the lesion. The beaded tungsten board and the robotic catheter are seen. The target and visual view are seen as well as representative images of the preplanning CT scan.

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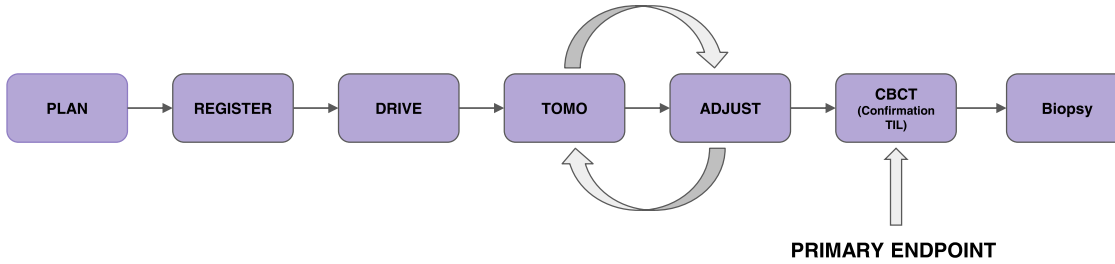


FIGURE 5. Study workflow. The procedure was planned and the airway was registered. In each case, the provider guided the robot to the virtual target, performed a TILT+ tomography (TOMO) sweep, and then corrected the position of the catheter for optimal alignment. A final TOMO sweep was performed to confirm the tool in the lesion, followed by a final CBCT confirmation spin.

center of the lesion. If the distance was >4 mm, repeat navigation was attempted at the discretion of the operator. Less than 4 mm was considered optimal for TILT+ confirmation (Fig. 6; an example of the Tomo reconstruction coordinate technique).

Statistical Analyses

Mean and standard deviation are reported for continuous variables; categorical variables are reported as percentages and counts. The statistical significance of differences among continuous variables was assessed using a *t* test. Two-tailed *P* values of ≤0.05 were considered statistically significant and analyses were performed using Google Sheets (Version 2022).

RESULTS

Lung nodule's average size was 16.3 ± 0.97 mm and were predominantly in the lower lobes (65%). Only 15% (3/20) had a bronchus sign and the average distance to the pleura was 6.88 ± 5.5 mm. All 4 operators successfully navigated to all (100%) of the lesions in an average of 3 minutes and 39 seconds. The average procedure time from navigation to the conclusion of biopsies was 32.45 minutes, ranging from 14 minutes to 87 minutes. The median number of tomosynthesis sweeps was 3 (range: 2 to 13), and augmented fluoroscopy was utilized in most cases (17/20 or 85%). Table 1 demonstrates baseline characteristics.

CBCT confirmation spins after the final TOMO sweep demonstratedTIL was 95% (19/20) with a center strike rate of 60% (12/20). One biopsy attempt was TTL (5%). Biopsy yielding purple pigmentation on microscopic or gross examination was 100% (20/20). Table 2 demonstrates navigation results.

DISCUSSION

Our study using electromagnetic-guided robotic bronchoscopy with digital tomosynthesis with TOMO reconstruction coordinate technique for TIL confirmation showed a 95% and TTL of

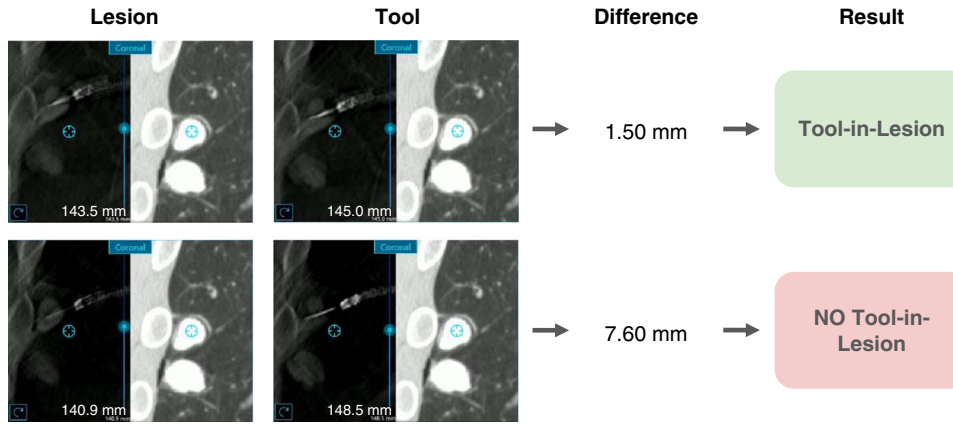


FIGURE 6. Tomosynthesis reconstruction coordinate technique top row: The lesion in the left lower lobe of the lung demonstrated an optimal image of the needle, the lung lesion of 1.5 mm, and was considered TIL. Bottom row: the distance between the optimal image of the lung nodule and the optimal image of the needle was 7.6 mm, which was >4 mm and was considered not a tool in the lesion.

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TABLE 1. Baseline Characteristics

	n (%)
Porcine weight (lbs), mean (range)	123 (117-127)
Nodule size (mm), mean ± SD	16.3 ± 0.97
Average minimum distance to pleura (mm), mean ± SD	6.88 ± 5.5
Bronchus sign	3 (15)
Nodule location	
Left lower lobe	9 (45)
Left middle lobe	1 (5)
Left upper lobe	1 (5)
Right lower lobe	4 (20)
Right middle lobe	1 (5)
Right upper lobe	4 (20)

5%. Although the intention of the biopsy was TIL confirmation, we had an incidental 60% center strike rate. Biopsy confirmation demonstrated diagnostic yield as defined the by the presence of intralesional purple pigment dye was 100% (20/20).

The ACCESS study utilizing Monarch in a cadaveric model utilized artificial tumor targets sized 10 to 30 mm in axial diameter and were implanted into 8 human cadavers. Sixty-seven nodules were evaluated in 8 cadavers. The mean nodule size was 20.4 mm. The overall diagnostic yield was 65/67 (97%).[15] The Precision-1 ION study in a cadaveric model demonstrated a rate of successful nodule localization and puncture of 20 pseudotumors was 80%.[16] The authors recognize that porcine and cadaveric experiments with highly controlled conditions typically outperform human trials. Conclusions as to the performance cannot be predicted based on solely animal or cadaveric trials alone. When comparing experiments, the authors believe cadaveric studies are less controlled than animal studies. Animal studies require anesthesia, prone to bleeding and atelectasis. Both Monarch and ION platforms performed similarly in human trials with high rates of lesion localization and lower rates of diagnostic yield. The Monarch BENEFIT multicenter prospective trial demonstrated that lesion localization was successful in 96.2% of patients but had a diagnostic yield of 74.1%.[17] Fielding et al[18] demonstrated an ION SS localization rate of 96% and a diagnostic yield of 79%.

TABLE 2. Navigation, TOMO, and CBCT Results

	n (%)
Time to a lesion, seconds (min)	219 (3.65)
Successful navigation to a lesion	20/20 (100)
Number of TOMO spins, median	3
Augmented fluoro used	17/20 (85)
Radiation per TOMO spin (mGy)	15.7
Radiation per CBCT spin (mGy)	46.3
TIL after the final TOMO	20/20 (100)
TIL confirmation by CBCT	20/20 (100)
Biopsy yield (purple pigmentation)	20/20 (100)

CBCT indicates cone-beam computerized tomography; TIL, tool-in-lesion; TOMO, tomography.

Monarch and ION robotic platforms utilize a preplanning CT scan to create an electronically generated virtual target. Electronically generated virtual targets are thought to be prone to CT2BD.[6] CT2BD can occur for various reasons, including atelectasis, neuromuscular weakness due to anesthesia, tissue distortion from the catheter system, bleeding, ferromagnetic interference, and perturbations in anatomy such as pleural effusions. Radial probe endobronchial ultrasound, often used to localize lesions and overcome CT2BD, has intrinsic issues as an intraoperative imaging device. REBUS is only lateral looking, unable to assess directionality with eccentric views, and prone to false positives due to atelectasis and focal hemorrhage. However, rEBUS can be helpful with concentric lesions with a higher rate of diagnostic yield.[10,14,19] CT2BD can increase the length of the procedure, frustrate the operator, and ultimately lead to a nondiagnostic procedure.[6] Despite the advancement of the first generation of robotic bronchoscopy systems, CT2BD limits improved diagnostic yield. Lesion localization although high in BENEFIT and Precision-1 at 96% may be potentially misleading to the reader and is only reflective of the catheter positioned at a virtual target. In contrast, the Galaxy System provides real-time intraprocedural imaging. In theory, the addition of digital tomosynthesis to a robotic platform may offer the provider TIL confirmation and improve provider confidence.

Digital tomosynthesis algorithms have been recently introduced for the correction of CT2BD. Pritchett et al,[20] in a 2-center trial, demonstrated superDimension Fluoroscopic Navigation system (Medtronic) improved 3-dimensional target overlap from 59.6% (28/47) to 83.0% (39/47) before and after location correction, respectively. A prospective single study center utilizing the first-generation LungVision system (Body Vision Medical Ltd.) demonstrated an average of 14.5 mm (range: 2.6 to 33.0 mm) from preprocedure CT to intraprocedural CBCT images. The average distance between the lesion location, as shown by LungVision augmented fluoroscopy

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system, and the actual location measured by CBCT was 5.9 mm (range: 2.1 to 10.0 mm).[20]

The Galaxy System hopes to improve diagnostic yield by combining integrated digital tomosynthesis and the advantages of a robotic catheter system. In the current study, TILT+ with TOMO reconstruction coordinate technique successfully corrected for CT2BD with a diagnostic yield of 100%. The authors agree that human trials are required to better assess the performance of the Galaxy System.

COMPLICATIONS

No significant complications occurred.

LIMITATIONS

There are several limitations. All 4 operators had significant bronchoscopy experience with digital TOMO and cone-beam CT imaging, limiting generalizability. The study findings may not apply to less experienced operators. Lesion characteristics (mixed gelatinous solution) do not reflect the various lesion characteristics encountered in clinical practice. The success rate with a porcine model does not necessarily indicate human success.

CONCLUSIONS

The Galaxy System demonstrated successful digital TOMO-confirmed TIL success in 95% (19/20) of lesions and TTL in 5% (1/20) as confirmed by CBCT. Successful diagnostic yield was achieved in 100% (20/20) of lesions as confirmed by intralesional pigment acquisition. Additional clinical trials are warranted to see whether high success rates can be reproduced in human trials.

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